



Corona: Two-Round Ring-LWE Threshold Signatures for Public Permissionless Chains

Zach Kelling

Lux Industries

2024

Abstract

We present Corona, a two-round Ring-LWE threshold signature scheme designed for public, permissionless blockchain consensus. Corona enables any t -of- n subset of a validator set to collaboratively sign blocks without any single party ever holding the full secret key, while providing security against quantum adversaries. The scheme operates in two rounds: a message-independent precomputation and commitment round, followed by a signing round, with a non-interactive combine producing the final aggregate signature. Corona builds on the Ring-LWE assumption over a single polynomial ring $R_q = \mathbb{Z}_q[X]/(X^N + 1)$, with Ring-SIS commitments. Ring-LWE is precisely Module-LWE specialised to module rank $k = 1$: the public key is a single ring relation $b = a \cdot s + e$ over R_q (with $a, s, e \in R_q$ and ring multiplication), rather than a matrix-vector product. Corona is the rank-1 base of the Lux lattice threshold family; Pulsar (LP-4450) is the rank- k Module-LWE generalisation. We prove that the scheme achieves existential unforgeability under chosen-message attacks (EUF-CMA) in the random oracle model, reducing security to the hardness of Ring-LWE together with Ring-SIS, at NIST security Category 1 (~ 128 -bit classical, ~ 130 -bit quantum). A Corona threshold signature is approximately 33,052 bytes and verifies in approximately 1.6 ms. Corona’s distinguishing contribution is making the academic two-round lattice threshold construction work for public, permissionless validator sets that change over time: leaderless distributed key generation with no trusted dealer (Feldman/Pedersen commitments), dynamic resharing across a changing validator set, and on-chain ceremony binding anchored to the Lux Q-Chain. It is the intra-lattice diversity leg of the Aurora certificate profile (Pulsar || Corona) under Quasar, verified by the raw EVM precompile at 0x012206. Corona extends the two-round lattice threshold construction of Boschini et al. (IACR ePrint 2024/1113) to the permissionless setting; it is a distinct production artifact.

Contents

1	Introduction	4
2	Preliminaries	5
2.1	The Cyclotomic Ring	5
2.2	Shamir Secret Sharing over R_q	5
2.3	Corona Parameters	5
3	The Corona Scheme	5
3.1	Distributed Key Generation (Leaderless, No Trusted Dealer)	6
3.2	Signing Protocol (2 Rounds)	6
3.3	Combine and Verification	7
4	Security Analysis	7
4.1	Unforgeability	7
4.2	Threshold Security	8
4.3	Concrete Security Estimates	8
5	Integration with Lux Quasar	9
5.1	On-Chain Verification	9
5.2	Permissionless Key Lifecycle	9
6	Empirical Evaluation	10
6.1	Benchmarks	10
6.2	Geographic Distribution	10
6.3	Comparison with Classical Threshold Schemes	10
7	Related Work	10

1 Introduction

Threshold signatures enable a group of n parties to collaboratively sign messages such that any subset of t or more parties can produce a valid signature, but fewer than t parties cannot. In blockchain consensus, threshold signatures aggregate validator votes into compact block certificates, reducing verification from $O(n)$ individual signature checks to a single aggregate verification.

Classical threshold schemes—BLS threshold [2], FROST [3], and CGGMP21 [4]—rely on discrete logarithm or pairing assumptions vulnerable to Shor’s algorithm [1]. Post-quantum threshold signatures based on lattice assumptions remain an active area of research. Boschini et al. [9] gave a practical two-round lattice threshold construction from LWE; that construction, however, assumes a fixed, known set of parties, which is precisely the assumption a public chain cannot make.

Corona is the first member of the Lux lattice threshold-signature family by development age: a two-round Ring-LWE threshold scheme together with its native verifier. Ring-LWE is Module-LWE specialised to module rank $k = 1$, so Corona works over a *single* polynomial ring $R_q = \mathbb{Z}_q[X]/(X^N + 1)$ with a single public ring element $a \in R_q$ and the relation $b = a \cdot s + e$, rather than a matrix $\mathbf{A} \in R_q^{M \times N'}$. Corona is the rank-1 base; Pulsar [12] (LP-4450) is the rank- k Module-LWE generalisation that lifts the same protocol from R_q to R_q^k . The design prioritizes:

- **Low round complexity:** 2 rounds (1 message-independent precomputation and commitment + 1 signing), then a non-interactive combine. Two rounds is optimal for any dealer-free threshold scheme.
- **Compact signature:** ~ 33 KB, acceptable for block certificates.
- **Fast verification:** ~ 1.6 ms (one ring relation check and one hash evaluation).
- **Standard assumptions:** Ring-LWE with Ring-SIS commitments, Category 1 parameters.
- **Permissionless operation:** leaderless distributed key generation with no trusted dealer, and dynamic resharing across a validator set that changes over time.

Corona’s distinguishing contribution is engineering, not a new hardness assumption: it takes the academic two-round lattice threshold structure and makes it run on a *public, permissionless* validator set. Where the academic construction fixes the participant set in advance, Corona supports validators joining and leaving permissionlessly through leaderless distributed key generation (Feldman/Pedersen commitments, no trusted dealer, no party ever holding the full key), dynamic resharing across generations, and on-chain ceremony binding anchored to the Lux Q-Chain.

Contributions.

1. The Corona two-round threshold signature scheme over Ring-LWE (rank-1) with Ring-SIS commitments (§3).
2. EUF-CMA security proof reducing to Ring-LWE and Ring-SIS hardness in the random oracle model (§4).
3. Permissionless, leaderless distributed key generation and resharing for validator sets that change over time, anchored to an on-chain ceremony root (§3, §5).
4. Concrete Category 1 parameters with security estimates (§4).
5. Integration as the Aurora-profile intra-lattice diversity leg under Lux Quasar (§5).
6. Benchmarks: ~ 33 KB signatures, ~ 1.6 ms verification (§6).

2 Preliminaries

2.1 The Cyclotomic Ring

Let $R = \mathbb{Z}[X]/(X^N + 1)$ and $R_q = \mathbb{Z}_q[X]/(X^N + 1)$ where N is a power of 2 and q is a prime with $q \equiv 1 \pmod{2N}$ (enabling the number-theoretic transform, NTT). Elements of R_q are polynomials of degree $< N$ with coefficients in $\mathbb{Z}_q$; the ring product is polynomial multiplication reduced modulo $X^N + 1$. For $f \in R_q$ we write $\|f\|_\infty$ for the largest absolute value of its coefficients (centred representatives in $(-q/2, q/2]$), and D_σ for the discrete Gaussian over R of width σ applied coefficient-wise. Corona operates over this *single* ring R_q ; all public values, secrets, errors, commitments, challenges and responses are elements (or short tuples) of R_q , never matrices over R_q .

Definition 2.1 (Ring-LWE). *The (decisional) Ring-LWE problem with parameters (N, q, σ) is: given a uniformly random $a \xleftarrow{\$} R_q$ and $b = a \cdot s + e$, where $s \xleftarrow{\$} D_\sigma$ and $e \xleftarrow{\$} D_\sigma$ are short and $\cdot$ is ring multiplication in R_q , distinguish (a, b) from (a, u) where $u \xleftarrow{\$} R_q$ [10].*

Remark 2.1 (Rank-1 specialisation of Module-LWE). *Ring-LWE is exactly Module-LWE specialised to module rank $k = 1$: the public matrix $\mathbf{A} \in R_q^{M \times N'}$ of the general module construction collapses to a single ring element $a \in R_q$, and the secret/error vectors collapse to single ring elements $s, e \in R_q$. Corona is the rank-1 base of the Lux lattice threshold family; Pulsar (LP-4450) is the rank- k Module-LWE generalisation, lifting R_q to R_q^k .*

Definition 2.2 (Ring-SIS). *The Ring-SIS problem with parameters (N, q, m, β) is: given uniformly random $a_1, \dots, a_m \xleftarrow{\$} R_q$, find $z_1, \dots, z_m \in R_q$, not all zero, with $\|z_i\|_\infty \leq \beta$ and $\sum_{i=1}^m a_i \cdot z_i = 0$ in R_q . Corona's commitments are binding under Ring-SIS, and a forgery yields a short Ring-SIS solution relative to a .*

2.2 Shamir Secret Sharing over R_q

The secret key is a single short ring element $s \in R_q$. It is shared with one degree- $(t - 1)$ polynomial over the ring.

Definition 2.3 (Ring-Based Shamir Sharing). *Given a secret $s \in R_q$, sample $a_1, \dots, a_{t-1} \xleftarrow{\$} R_q$ and form $f(X') = s + \sum_{i=1}^{t-1} a_i (X')^i \in R_q[X']$, evaluated at distinct public points $X' = \text{id}_i \in \mathbb{Z}_q$ (the indeterminate X' of the sharing polynomial is distinct from the ring indeterminate X). Party i receives the share $s_i = f(\text{id}_i) \in R_q$.*

Lagrange interpolation over R_q reconstructs s from any t shares:

$$s = \sum_{i \in S} \lambda_i \cdot s_i, \quad \lambda_i = \prod_{j \in S, j \neq i} \frac{\text{id}_j}{\text{id}_j - \text{id}_i} \in \mathbb{Z}_q \quad (1)$$

where the Lagrange coefficients λ_i are scalars in $\mathbb{Z}_q$ (embedded in R_q).

2.3 Corona Parameters

3 The Corona Scheme

Corona is a two-round Ring-LWE threshold signature over the single ring R_q . The public key is one ring relation $b = a \cdot s + e$ with $a, s, e \in R_q$; there is no matrix. We first describe the leaderless distributed key generation (no trusted dealer), then the two-round signing protocol, then the non-interactive combine and verification. Throughout, $\mathcal{S}$ denotes a signing set with $|\mathcal{S}| \geq t$, and $\lambda_i \in \mathbb{Z}_q$ are the Lagrange coefficients of (1) for the set $\mathcal{S}$.

Parameter	Symbol	Value
Ring dimension	N	1024
Modulus	q	12289 ($q \equiv 1 \pmod{2N}$, NTT-friendly)
Module rank	k	1 (Ring-LWE)
Gaussian width (key gen)	σ_e	3.192
Gaussian width (signing)	σ^*	$\gg \kappa \cdot \ s\ _\infty$
Small-element bound	β	32
Challenge ternary weight	κ	60

Table 1: Corona parameters (NIST Category 1, Ring-LWE rank-1). The single polynomial ring and rank $k = 1$ distinguish Corona from the rank- k Pulsar generalisation; the detailed sampler and rejection bounds are fixed by the reference implementation and its KAT vectors [14].

3.1 Distributed Key Generation (Leaderless, No Trusted Dealer)

The public ring element $a \in R_q$ is a fixed, publicly derivable parameter (a nothing- up-my-sleeve hash output bound to the ceremony root), so no party controls it. The secret $s \in R_q$ is generated jointly: each party contributes an additive summand and a verifiable Shamir sharing of that summand, so the group secret is $s = \sum_i s^{(i)}$ and no party ever sees s .

Algorithm 1 Corona Distributed Key Generation

Require: Threshold t , parties n , parameters (N, q, σ_e) , ceremony root ρ

- 1: $a \leftarrow H_{R_q}(\text{"Corona-pk"} \parallel \rho)$ $\triangleright$ public ring element, derived (no dealer)
- 2: **for** each party $i \in \{1, \dots, n\}$ **do** $\triangleright$ run in parallel; leaderless
- 3: $s^{(i)} \xleftarrow{\$} D_{\sigma_e}, \quad e^{(i)} \xleftarrow{\$} D_{\sigma_e}$ $\triangleright$ i 's short secret/error summands $\in R_q$
- 4: $f^{(i)}(X') \leftarrow s^{(i)} + \sum_{\ell=1}^{t-1} a_\ell^{(i)} (X')^\ell, \quad a_\ell^{(i)} \xleftarrow{\$} R_q$ $\triangleright$ Shamir poly over R_q
- 5: Broadcast Feldman/Pedersen commitments to $s^{(i)}$ and $\{a_\ell^{(i)}\}$
- 6: Send share $s_j^{(i)} \leftarrow f^{(i)}(\text{id}_j)$ privately to each party j
- 7: **end for**
- 8: **for** each party j **do** $\triangleright$ verify and aggregate
- 9: **abort/identify** any party i whose share $s_j^{(i)}$ contradicts its commitments
- 10: $s_j \leftarrow \sum_i s_j^{(i)}$ $\triangleright$ j 's aggregate share of $s = \sum_i s^{(i)}$
- 11: **end for**
- 12: $b \leftarrow a \cdot s + e, \quad e = \sum_i e^{(i)}$ $\triangleright$ public key, recombined from commitments
- 13: **return** $\text{pk} = (a, b), \quad \text{shares } \{s_j\}_{j=1}^n$ $\triangleright$ group secret s never reconstructed

The public key $b = a \cdot s + e$ is a Ring-LWE sample (Definition 2.1): recovering s from (a, b) is the Ring-LWE search problem. Because a is derived from the ceremony root and every contribution is committed, the protocol is leaderless and robust against an adversary controlling up to $t - 1$ parties; corrupted shares are caught by commitment verification (identifiable abort). Resharing for a changed validator set re-runs Algorithm 1 over the new set, with the group public key b held fixed across generations (see §5).

3.2 Signing Protocol (2 Rounds)

Round 1 is message-independent and can be precomputed. Each signer commits to a short masking element $y_i \in R_q$ via $w_i = a \cdot y_i$; correlated randomness with pairwise cancellation keeps the aggregate mask short, and per-message MACs authenticate commitments against malicious injection.

Algorithm 2 Corona Signing — Round 1 (Message-Independent)

Require: Party i , signers set $\mathcal{S}$ with $|\mathcal{S}| \geq t$, session id sid

- 1: $r_i \leftarrow \text{PRF}(\text{key}_i, \text{sid} \| i) \bmod D_{\sigma^*}$ $\triangleright i$'s short mask $\in R_q$
 - 2: **for** each party $j \in \mathcal{S} \setminus \{i\}$ **do**
 - 3: $r_{i,j} \leftarrow \text{PRF}(\text{seed}_{i,j}, \text{sid})$ $\triangleright$ pairwise correlated randomness $\in R_q$
 - 4: **end for**
 - 5: $y_i \leftarrow r_i + \sum_{j:j>i} r_{i,j} - \sum_{j:j<i} r_{j,i}$ $\triangleright$ masked randomness, pairwise cancellation
 - 6: $w_i \leftarrow a \cdot y_i$ $\triangleright$ commitment (single ring multiplication)
 - 7: **for** each $j \in \mathcal{S} \setminus \{i\}$ **do**
 - 8: $\text{mac}_{i \rightarrow j} \leftarrow \text{HMAC}(\text{MACkey}_{i \rightarrow j}, \text{sid} \| w_i)$
 - 9: **end for**
 - 10: **broadcast** $(w_i, \{\text{mac}_{i \rightarrow j}\}_{j \in \mathcal{S}})$
-

Algorithm 3 Corona Signing — Round 2 (Message-Dependent)

Require: Party i , message μ , Round 1 data $\{(w_j, \text{mac}_j)\}_{j \in \mathcal{S}}$

- 1: **for** each $j \in \mathcal{S} \setminus \{i\}$ **do**
 - 2: Verify $\text{mac}_{j \rightarrow i}$ using $\text{MACkey}_{j \rightarrow i}$; **abort** (identify j) if it fails
 - 3: **end for**
 - 4: $w \leftarrow \sum_{j \in \mathcal{S}} \lambda_j \cdot w_j$ $\triangleright$ combined commitment $\in R_q$
 - 5: $c \leftarrow H(a, b, w, \mu)$ $\triangleright$ challenge: ternary polynomial in R_q , weight κ
 - 6: $z_i \leftarrow y_i + c \cdot \lambda_i \cdot s_i$ $\triangleright$ partial response (ring multiplication)
 - 7: **if** $\|z_i\|_\infty > B$ **then** $\triangleright$ Fiat–Shamir-with-aborts rejection sampling
 - 8: **abort** and restart with a fresh session id
 - 9: **end if**
 - 10: **broadcast** z_i
-

3.3 Combine and Verification

The combine is non-interactive: any party or external aggregator sums the partial responses into a single ring element

$$z = \sum_{i \in \mathcal{S}} \lambda_i \cdot z_i \in R_q,$$

and outputs the signature (c, z) . By linearity and the Lagrange reconstruction of (1), $z = y + c \cdot s$ where $y = \sum_i \lambda_i y_i$ masks the short secret s , so $a \cdot z - c \cdot b = a \cdot y - c \cdot e \approx w$.

Verification of (c, z) against $\text{pk} = (a, b)$ and message μ checks:

1. $\|z\|_\infty \leq B'$ (the response is short, for the combined bound B');
2. recompute $w' \leftarrow a \cdot z - c \cdot b \in R_q$ and check $c = H(a, b, w', \mu)$.

Both checks are single ring operations over R_q plus one hash, giving the ~ 1.6 ms verification of §6. A successful forgery on a fresh message yields a short z^* with $a \cdot z^* \approx w^* + c^* \cdot b$, i.e. a short Ring-SIS solution relative to a (Definition 2.2); this is the basis of the reduction in §4.

4 Security Analysis

4.1 Unforgeability

Theorem 4.1 (EUF-CMA Security). *Corona is existentially unforgeable under chosen-message attacks in the random oracle model, assuming the hardness of Ring-LWE(N, q, σ_e) (Definition 2.1) and Ring-SIS(N, q, m, β) (Definition 2.2). Concretely, any adversary making Q_H*

hash queries and Q_S signing queries has advantage:

$$\text{Adv}^{\text{EUF-CMA}} \leq \text{Adv}^{\text{R-LWE}} + Q_H \cdot \text{Adv}^{\text{R-SIS}} + \frac{Q_S}{2^{128}}.$$

Proof. We proceed through a sequence of games.

Game 0: The real EUF-CMA game. The adversary controls $t - 1$ parties and interacts with the signing oracle.

Game 1: Replace the public key $b = a \cdot s + e$ with a uniformly random $u \xleftarrow{\$} R_q$. The distinguishing advantage between Games 0 and 1 is $\text{Adv}^{\text{R-LWE}}$ by Definition 2.1: the pair (a, b) is exactly a Ring-LWE sample and (a, u) its uniform counterpart.

Game 2: In Game 1 the public key $b = u$ is independent of s . We reprogram the random oracle H to extract the forgery. When the adversary outputs a forgery (c^*, z^*) on a fresh μ^* , validity requires

$$a \cdot z^* - c^* \cdot u \approx w^* \quad \text{with } c^* = H(a, u, w^*, \mu^*),$$

i.e. a short z^* with $a \cdot z^* \approx w^* + c^* \cdot u$. Since u is uniform and independent of s , producing such a short z^* is an instance of Ring-SIS (Definition 2.2) relative to a : it yields a short nonzero ring combination summing to zero. Each of the Q_H oracle queries gives one such extraction opportunity, whence the $Q_H \cdot \text{Adv}^{\text{R-SIS}}$ term.

The probability of a challenge collision in H (two queries yielding the same ternary challenge for different messages) is $\leq Q_H^2/|C|$, where the challenge space $|C| = \binom{N}{\kappa} 2^\kappa$ is negligible for $\kappa = 60$ and $N = 1024$.

Combining the transitions, $\text{Adv}^{\text{Game 0}} \leq \text{Adv}^{\text{R-LWE}} + Q_H \cdot \text{Adv}^{\text{R-SIS}} + Q_H^2/|C|$. $\square$

4.2 Threshold Security

Lemma 4.2 (Threshold Unforgeability). *An adversary controlling $t - 1$ parties and observing Q_S signing sessions cannot forge a signature on a new message. The adversary's view in each signing session is statistically close to a simulation that does not use the honest parties' shares, by the zero-knowledge property of the commitment scheme.*

Proof. In each signing session, the adversary sees the honest parties' commitments $w_i = a \cdot y_i$ and responses $z_i = y_i + c\lambda_i s_i$. Given w_i and z_i , the adversary can compute $c\lambda_i s_i = z_i - y_i$. However, reconstructing y_i from $w_i = a \cdot y_i$ is an instance of Ring-SIS (finding a short ring preimage of w_i relative to a , Definition 2.2).

The Fiat–Shamir-with-aborts rejection sampling ensures that z_i is statistically independent of s_i : the distribution of z_i conditioned on any s_i is within statistical distance 2^{-128} of the distribution conditioned on any other s_i , by the properties of discrete Gaussian rejection sampling with width $\sigma^* \gg \|c\lambda_i s_i\|_\infty$.

Therefore $t - 1$ parties' shares reveal no information about the group secret s (by the Shamir sharing of Definition 2.3), and the signing transcripts reveal no additional information about s (by the zero-knowledge simulation). The same argument applies to the distributed key generation: an adversary corrupting up to $t - 1$ parties learns nothing about $s = \sum_i s^{(i)}$ beyond what the public key b already reveals, since the honest contributions $s^{(i)}$ remain hidden behind their commitments. $\square$

4.3 Concrete Security Estimates

Proposition 4.3 (Security Level). *With the Category 1 parameters $N = 1024$, $q = 12289$, rank $k = 1$, $\sigma_e = 3.192$ (Table 1), the Ring-LWE instance reaches NIST security Category 1: ~ 128 -bit classical and ~ 130 -bit quantum security against the best known lattice algorithms (BKZ with sieving, Grover-enhanced sieving).*

Proof. Ring-LWE at rank 1 embeds into an LWE lattice of dimension $d = N = 1024$ (the rank-1 specialisation; a rank- k module would give $d = kN$). The primal lattice attack via BKZ with sieving has classical cost

$$T_{\text{classical}} \approx 2^{0.292\beta + o(\beta)},$$

where the blocksize β is the smallest value at which BKZ recovers the secret for the (d, q, σ_e) instance; for these parameters β places the cost at the Category 1 level, $T_{\text{classical}} \approx 2^{128}$. For quantum attacks (Grover-accelerated sieving) the exponent constant drops to 0.265, giving $T_{\text{quantum}} \approx 2^{130}$ at the same blocksize. The Ring-SIS commitments are parameterised over the same ring and modulus and are no easier than the Ring-LWE instance. Corona is therefore a Category 1 (Ring-LWE) scheme. It is *not* byte-equal to any FIPS standard—byte-equality with FIPS 204 ML-DSA is a property of the rank- k Pulsar generalisation (LP-4450), not of rank-1 Corona. $\square$

5 Integration with Lux Quasar

Corona is composed under the Lux Quasar consensus engine as the intra-lattice diversity leg of multi-leg finality certificates. A Quasar certificate profile bundles independently verifiable signature legs, each of which must validate for finality:

1. **BLS leg:** classical BLS12-381 threshold signatures provide the fast (1-round) aggregation lane.
2. **Pulsar leg:** threshold ML-DSA over Module-LWE (rank- k , FIPS 204 byte-equal, LP-4450) is the post-quantum floor.
3. **Corona leg:** this scheme—two-round Ring-LWE (rank-1)—is the intra-lattice diversity leg, profile-selectable alongside Pulsar.

The **Aurora** certificate profile (LP-4900) is Pulsar $\parallel$ Corona; the **Polaris** profile (LP-4910) is Pulsar $\parallel$ Corona $\parallel$ Magnetar, adding the hash-based Magnetar leg (SLH-DSA, LP-4540).

Diversity caveat. Corona and Pulsar are *both* lattice constructions: Corona is the rank-1 Ring-LWE base and Pulsar the rank- k Module-LWE generalisation. The Pulsar $\parallel$ Corona composition of Aurora is therefore *intra-lattice* defence-in-depth—different code paths, different parameter sets, different rejection samplers—and is honestly *not* family-disjoint hardness: a structural break of the cyclotomic-ring lattice problem could in principle touch both legs. True cross-family diversity is provided by the hash-based Magnetar leg in the Polaris profile, whose security rests on hash (SLH-DSA, FIPS 205) rather than lattices. Aurora buys implementation and parameter diversity; Polaris buys cryptographic-family diversity.

5.1 On-Chain Verification

A Corona threshold signature is verified by the native EVM precompile at the raw slot 0x012206 (LP-0120): the precompile checks the aggregated two-round signature against the group public key in ~ 1.6 ms. Under the P3Q unified rollup-batch verifier (LP-218 / LP-220) Corona is **kind byte** 0x02 at slot 0x012205; the raw 0x012206 path remains for non-rollup-batch use.

5.2 Permissionless Key Lifecycle

Corona’s distributed key generation (Algorithm 1) runs leaderless over the *current* validator set, with no trusted dealer, anchored to the Q-Chain ceremony root ρ . As validators join and leave permissionlessly, the key material is reshared across generations: each generation re-runs the DKG over the new set while holding the group public key fixed for cross-generation verification, and snapshots are committed on-chain. Rotation is rate-limited (a minimum dwell between rotations) and force-rotated on an upper bound, so a churning validator set cannot

exhaust the rotation machinery. This permissionless, on-chain-anchored lifecycle—rather than any change to the underlying two-round lattice algebra—is what adapts the academic threshold construction to a public chain.

6 Empirical Evaluation

6.1 Benchmarks

Operation	Time	Size
Distributed key generation (3 parties)	1.2s	— (no dealer; shares local)
Round 1 (per party, precomputable)	85ms	commitment $w_i \in R_q$
Round 2 (per party)	120ms	response $z_i \in R_q$
Total signing (2-of-3)	0.6s	—
Verification	1.6ms	—
Signature size	—	$\sim 33,052$ B

Table 2: Corona performance benchmarks (Ring-LWE rank-1, Category 1; signature size and verification time per LP-0120).

6.2 Geographic Distribution

With 5 parties across US-East, US-West, EU-West, Asia-Pacific, and South America (100–250 ms inter-region latency), end-to-end signing for a 3-of-5 threshold was dominated by network round-trips rather than computation, comfortably within the finality-certificate cadence. Because Round 1 is message-independent, it can be precomputed, leaving only the Round 2 round-trip and the non-interactive combine on the critical path.

6.3 Comparison with Classical Threshold Schemes

Scheme	Sig Size	Sign Time	Verify	PQ Safe
BLS Threshold	48 B	5ms	2ms	No
FROST	64 B	10ms	1ms	No
CGGMP21	65 B	200ms	<1ms	No
Corona	~ 33 KB	600ms	1.6ms	Yes

Table 3: Threshold signature scheme comparison. Corona is the price of quantum resistance on the same dealer-free, two-round profile as FROST.

Corona’s ~ 33 KB signature is larger than classical schemes by roughly three orders of magnitude, but this is acceptable for block certificates (one signature per block). As the rank-1 base, Corona is more compact than the rank- k Pulsar generalisation, since a single ring element replaces a length- k vector of ring elements. The signing time is dominated by network latency rather than computation, and the ~ 1.6 ms verification is competitive with classical schemes.

7 Related Work

Bendlin et al. [5] introduced lattice-based threshold signatures using NTRU. Boneh et al. [6] proposed threshold signatures from lattice assumptions but with impractical parameters. Cozzo

and Smart [7] and Espitau et al. [8] improved practical lattice-based threshold schemes. Closest to Corona, Boschini et al. [9] gave a practical *two-round* threshold signature from LWE; their construction assumes a fixed, known set of parties. Corona’s two-round structure follows that line, and Corona is Lux’s own production scheme: it extends the construction to public, permissionless validator sets via leaderless distributed key generation, dynamic resharing, and on-chain ceremony binding—engineering that the academic construction does not provide. The Ring-LWE ring R_q itself is the ideal-lattice setting of Lyubashevsky, Peikert and Regev [10].

Within the Lux lattice threshold family, Corona is the rank-1 Ring-LWE base. Pulsar [12] (LP-4450) is the rank- k Module-LWE generalisation—threshold ML-DSA-65, byte-equal to FIPS 204—which lifts Corona’s single-ring algebra to polynomial vectors over R_q^k . Magnetar [13] (LP-4540) is the hash-based threshold leg (SLH-DSA, FIPS 205), providing the cross-family diversity that two lattice schemes cannot. Corona, Pulsar and Magnetar are composed by the Aurora and Polaris certificate profiles under Quasar.

8 Conclusion

Corona is the foundational member of the Lux lattice threshold-signature family: a two-round Ring-LWE (rank-1) threshold scheme over a single polynomial ring R_q , with Ring-SIS commitments and Category 1 security (~ 128 -bit classical, ~ 130 -bit quantum). It produces ~ 33 KB signatures that verify in ~ 1.6 ms via the native precompile at `0x012206`. Its distinguishing contribution is engineering rather than a new hardness assumption: Corona takes the academic two-round lattice threshold construction [9] and makes it run on a public, permissionless validator set through leaderless distributed key generation with no trusted dealer, dynamic resharing across a changing set, and on-chain ceremony binding anchored to the Lux Q-Chain. Under Quasar, Corona is the intra-lattice diversity leg of the Aurora profile (Pulsar || Corona) and Polaris profile (Pulsar || Corona || Magnetar); the rank- k Module-LWE generalisation is Pulsar [12] (LP-4450), and the cross-family hash leg is Magnetar [13] (LP-4540). The reference implementation is `github.com/luxfi/corona` [14], with machine-checked correctness in `proofs/lean/Crypto/Corona.lean`.

References

- [1] P. Shor, “Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer,” *SIAM J. Computing*, 1997.
- [2] A. Boldyreva, “Threshold signatures, multisignatures and blind signatures based on the gap-Diffie-Hellman-group signature scheme,” in *PKC*, 2003.
- [3] C. Komlo and I. Goldberg, “FROST: Flexible round-optimized Schnorr threshold signatures,” in *SAC*, 2020.
- [4] R. Canetti et al., “UC non-interactive, proactive, threshold ECDSA with identifiable aborts,” in *CCS*, 2021.
- [5] R. Bendlin, S. Krehbiel, C. Peikert, and A. Rosen, “Semi-homomorphic encryption and multiparty computation,” in *EUROCRYPT*, 2013.
- [6] D. Boneh et al., “Threshold cryptosystems from threshold fully homomorphic encryption,” in *CRYPTO*, 2018.
- [7] D. Cozzo and N. Smart, “Sharing the LUOV: Threshold post-quantum signatures,” in *IMA*, 2023.

- [8] T. Espitau, M. Tibouchi, and A. Wallet, “Practical Fiat-Shamir with aborts threshold signatures from lattices,” 2024.
- [9] G. Boschini, A. Kaviani, R. W. F. Lai, G. Malavolta, A. Takahashi, and M. Tibouchi, “Ringtail: Practical two-round threshold signatures from LWE,” IACR ePrint 2024/1113; IEEE S&P 2025.
- [10] V. Lyubashevsky, C. Peikert, and O. Regev, “On ideal lattices and learning with errors over rings,” in *EUROCRYPT*, 2010.
- [11] O. Regev, “On lattices, learning with errors, random linear codes, and cryptography,” *JACM*, 2009.
- [12] Lux Core Team, “Pulsar: Module-LWE (rank- k) threshold ML-DSA-65 (FIPS 204 byte-equal),” Lux Proposal LP-4450.
- [13] Lux Core Team, “Magnetar: hash-based threshold SLH-DSA (FIPS 205),” Lux Proposal LP-4540.
- [14] Lux Core Team, “Corona reference implementation,” github.com/luxfi/corona (v0.7.6); Lux Proposal LP-4440.